

Analysis of Uncertainty Influence on the Probabilistic Evaluation of Vortex-induced Vibration Response of a Self-anchored Suspension Bridge

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Abstract

The uncertainty of the parameters involved in the bridge vortex-induced vibration (VIV) response is hard to be described accurately and the probabilistic description remains scarce. The present work aims at giving a contribution to the probabilistic evaluation of the bridge VIV response considering uncertainty influence of parameters. A self-anchored suspension bridge was selected as an example to analyze the uncertainty influence on the probabilistic evaluation of VIV response. The probability density function of the wind speed at the deck level of the bridge was established. The modal analysis was carried out and the natural vibration characteristics were checked. Then three performance functions involving different considerations of the maximum response models of bridge VIV were introduced. The numerical characteristics (coefficient of variation (COV), skewness and kurtosis) in the parameter space was propagated to access its influence on the exceeding probability of the bridge VIV response utilizing the second-order fourth-moment (SOFM) method. The influence from the different distribution type of the parameters was investigated. Results showed that the influence degree of COV, skewness and kurtosis of parameters on the exceeding probability of VIV response for the bridge is different and the distribution type of parameter has little effect based on the SOFM method.

Keywords: bridge VIV response, performance function, uncertainty, numerical characteristics, exceeding probability

1. Introduction

Estimation of the aeroelastic stability is a fundamental part of the structural designs. Corresponding studies can be performed by experimental tests in a wind tunnel or by a hybrid methodology that consists of wind tunnel tests and subsequent computation stages. The aim in both cases is to determine the wind loading and the extreme response for structures or to find out the critical wind speed value of static instability, flutter or galloping especially for bridges. On the other hand, some empirical formulas expressed in terms of structural properties, basic frequencies and aerodynamic coefficients, have been developed in order to provide estimate values of aeroelastic behavior. Although these formulas are used in design codes of different countries, it is known that in fact these values are not accurate enough because of the model error and specific wind tunnel tests should be performed for each case. And many parameters used for estimation of aeroelastic response are associated with a number of uncertainties. It is necessary to use a probabilistic approach to include this consideration about uncertainties of parameters in the wind-excited structural response analysis.

Some relevant works related to the reliability analysis of wind-excited vibration about flutter (Argentini *et al.*, 2014; Baldomir *et al.*, 2013; Canor *et al.*, 2015; Cheng *et al.*, 2005; Ge *et al.*,

2000; Jurado and Hernández, 2004; Kusano *et al.*, 2014; Ostenfeld-Rosenthal *et al.*, 1992; Pourzeynali and Datta, 2002; Seo and Caracoglia, 2011; Zasso *et al.*, 1996), vortex-induced vibration (VIV) (Ge *et al.*, 2004; Xiang *et al.*, 2005) and buffeting (Caracoglia, 2008; Cheng and Li, 2009; Hu and Ou, 2013; Pourzeynali and Datta, 2005; Seo and Caracoglia, 2012; Seo and Caracoglia, 2013; Su *et al.*, 2010; Wu *et al.*, 2012; Zhao and Ge, 2013) were conducted using the first-order second-moment (FOSM), second-order second-moment (SOSM), peak over threshold or stochastic finite element methods. The random nature associated with natural frequency, structural damping, wind load factor was assessed by incorporating these random variables into the complex-valued eigenvalue or eigenvector system through a number of statistic features with an approximation of the limit state function or risk evaluation of fatigue failure. It is also necessary to consider the uncertainty influence of aerostatic/aerodynamic coefficients affecting the structural wind-induced vibration response.

Bridge VIV is a limited amplitude vibration that will not cause collapse of structure, but it can result in large displacements and hence uncomfortable issues. Since bridge VIV commonly occurs in the low wind speed region with a relatively high occurrence probability, it may lead to a long-term fatigue damage. Therefore, it is important to assess the exceeding probability of the bridge VIV response. However, deterministic analyses of VIV for

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bridges have been carried out and a few semi-empirical vortex-induced force models (Diana *et al.*, 2006; Ehsan and Scanlan (1990); Ehsan *et al.*, 1990; Larsen, 1995; Marra *et al.*, 2011; Mashnad and Jones, 2014; Scanlan, 1998; Wu and Kareem, 2013; Xu *et al.*, 2015) have been proposed based on the assumption of complete determinacy of structural parameters. The investigation about reliability analysis of VIV is few. Ehsan and Scanlan (Ehsan and Scanlan (1990)) analyzed the variability of fitting parameters in Van der Pol nonlinear model for VIV. Prenninger (Prenninger *et al.*, 1990) considered the effect of VIV caused by high order mode of cable-stayed bridge on wind-resistant reliability in reliability analysis and it demonstrated that VIV had great influence. Ge *et al.* (2004) assessed the probability for VIV of an arch bridge and its essence was occurrence probability of lock-in wind speed. Li and Liao (Li and Liao, 2003) carried out the dispersing of time and space for vortex-induced force by use of strip hypothesis in conjunction with the second Scanlan's vortex shedding half-empirical and half-analytical model and an Alternating Time Frequency (AFT) domain method was developed to solve the calculation of vortex-induced responses.

In this study, a self-anchored suspension bridge was presented to analyze the influence of parametric uncertainties resulting from both bridge properties and aerodynamic on the VIV response based on the second-order fourth-moment (SOFM) method, which is verified by using the Monte Carlo simulation (MCS) methods. The SOFM method retains the linear and quadratic parts of Taylor series expansion of the performance function. Moreover, this method involves the information of the first four statistical moments of the random variables, of which the distribution types can be unknown. A procedure based on the MCS method was implemented to allow for the direct numerical estimation of the exceeding probability as algorithm verification of the SOFM method. Three performance functions involving different considerations of the maximum response models of bridge VIV were introduced. The exceeding probabilities of the bridge VIV response and the influence of the various random nature on the exceeding probabilities were assessed based on various reliability analysis approaches.

2. A Self-anchored Suspension Bridge

A self-anchored suspension bridge with single tower in China was analyzed as a numerical example (Li *et al.*, 2019). The main

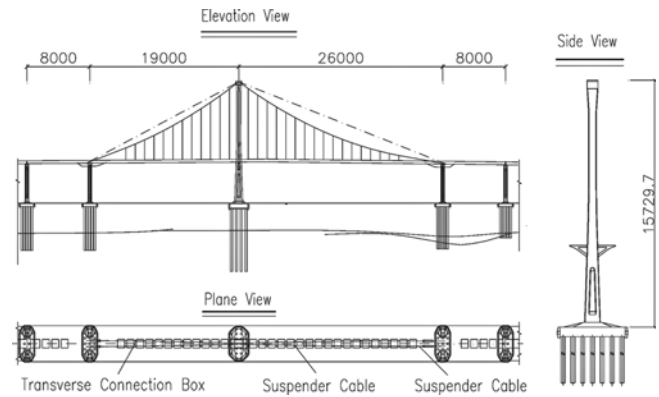


Fig. 1. General Layout of the Self-anchored Suspension Bridge (unit: cm)

span, total span length and tower height of the bridge is 260 m, 610 m and 157 m separately, as shown in Fig. 1. The bridge deck is a central-slotting steel box girder with crossbeam connection. The total width, slot width and height of the typical cross section are 47 m, 11 m and 3.6 m respectively, as shown in Fig. 2. The damping ratio of the bridge deck is 0.5%.

2.1 Distribution of Wind Velocity

Although the wind velocity data at the bridge site is hard to be obtained due to the high cost and shorter sampling time, the distribution of wind velocity can be estimated by utilizing the data of nearby meteorological stations and wind-resistance specifications. For the wind velocity statistics, the yearly maximum wind velocity is used as the sample data by various countries' standards. As is well known, the cumulative distribution of the wind velocity tends to fit the asymptotic extreme-value distributions such as the extreme value type I (i.e., the Gumbel distribution), the extreme value type II (i.e., the Frechet distribution), and the extreme value type III (i.e., the Weibull distribution) (Mayne, 1979; Thoft-Christensen and Baker, 1982; Palutikof *et al.*, 1999). In this study, the distribution of wind velocity at bridge site was obtained based on the nearby meteorological station and the Chinese Standard (2004).

Based on the sampling data of the daily maximum wind velocity (10-min average time interval) at the meteorological station, the yearly 10-min average wind velocity (AWV) and the yearly maximum wind velocity (MWV) for a period of 40 consecutive years at 10 m above ground level height were

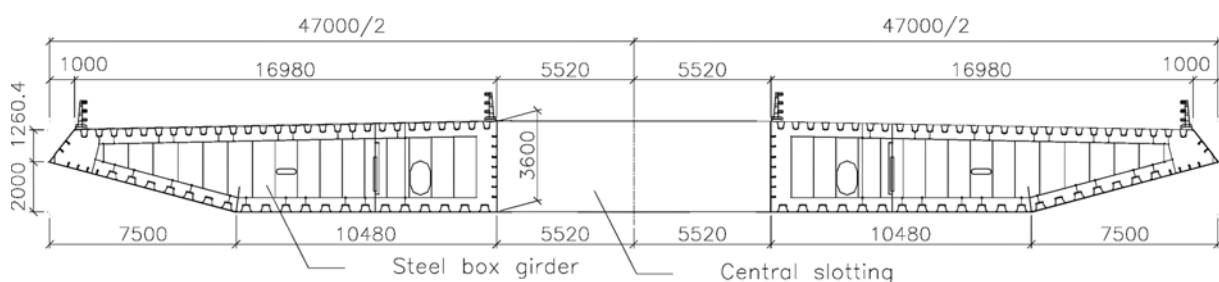


Fig. 2. Typical Cross Section of the Bridge Deck (unit: mm)

Table 1. Yearly Wind Velocity at the Meteorological Station

Year	AWV	MWV	Year	AWV	MWV
1970	2.729	14.523	1990	3.081	14.347
1971	2.729	12.323	1991	2.993	14.699
1972	2.817	12.323	1992	3.169	11.619
1973	2.641	11.443	1993	3.081	11.971
1974	2.729	12.323	1994	3.257	12.323
1975	2.465	10.562	1995	3.169	13.291
1976	3.785	16.724	1996	3.081	14.347
1977	3.873	16.724	1997	3.169	11.795
1978	3.521	18.484	1998	3.081	11.619
1979	3.169	15.844	1999	3.081	13.643
1980	3.345	15.844	2000	2.905	12.147
1981	3.257	15.844	2001	2.817	12.147
1982	3.169	15.844	2002	2.817	13.203
1983	3.609	16.460	2003	2.993	13.819
1984	3.345	16.724	2004	2.641	11.531
1985	3.169	17.604	2005	2.641	13.643
1986	3.169	14.083	2006	2.377	12.059
1987	3.257	16.460	2007	2.112	9.858
1988	2.993	12.323	2008	2.201	10.474
1989	3.081	14.083	2009	2.024	11.091

obtained. The results of the yearly 10-min AWV and MWV and the frequency statistics of the daily maximum wind velocity are shown in Table 1 and Table 2. The ground roughness exponent α was 0.16. By extreme wind fitting, the extreme value type I (i.e., Gumbel distribution) is fitted the sample data well and the cumulative distribution function (CDF) and the probability density function (PDF) is expressed as:

$$\text{CDF: } F(x) = \exp \left\{ -\exp \left[-\left(\frac{x-b}{a} \right) \right] \right\} \quad (1)$$

$$\text{PDF: } f(x) = \frac{1}{a} \exp \left\{ -\left(\frac{x-b}{a} \right) - \exp \left[-\left(\frac{x-b}{a} \right) \right] \right\} \quad (2)$$

where a and b are the scale parameter and location parameter, respectively.

The least absolute deviation (LAD) method was utilized to fit the parameters in the distribution model of extreme value type I. Three optimality indexes of parameter estimation were chosen to evaluate the results, shown as:

$$\text{Residual variance: } S_f^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \hat{x}_i)^2 \quad (3)$$

$$\text{Fitting relative deviation: } R_f = \frac{1}{n} \sum_{i=1}^n \frac{|x_i - \hat{x}_i|}{x_i} \quad (4)$$

Table 2. Frequency Statistics of Wind Velocity at the Station

Wind speed range /(m/s)	0~1	1~2	2~3	3~4	4~5	5~6	6~7	7~8	8~9
Frequency /%	0.0	0.1	6.1	16.7	19.3	17.9	15.3	12.0	6.5
Wind speed range /(m/s)	8~10	10~11	11~12	12~13	13~14	14~15	15~16	16~17	17~18
Frequency /%	3.1	1.6	0.8	0.3	0.2	0.1	0.0	0.0	0.0

Table 3. Wind Distribution Parameters and Optimality Index Evaluation

Parameter estimation methods	a	b	$S_f^2 / (\text{m}^2/\text{s}^2)$	$R_f / \%$	K_f
LAD	1.76	4.733	0.101	2.117	1.21

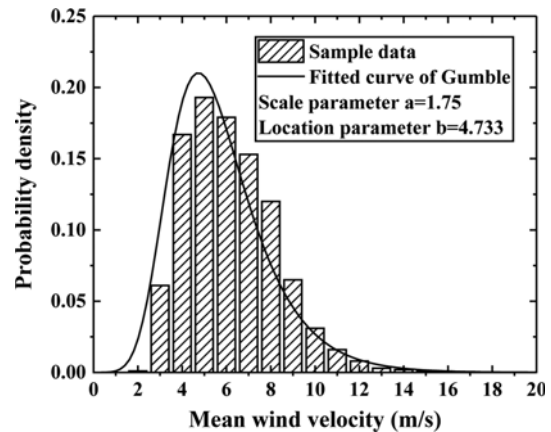


Fig. 3. Probability Density Curve of the Meteorological Station at 10 m Height

Appropriate test index of Kolmogorov fitting: $K_f = D_n \sqrt{n}$ (5)

$$D_n = \max \{ |F^*(x_i) - F(x_i)| \} \quad (6)$$

where n is the sample size; x_i ($i = 1, 2, \dots, n$) is the ordinal sample; $x_1 \leq x_2 \leq \dots \leq x_n$; $\hat{x}_i$ ($\hat{x}_i = b - a \ln[-\ln F^*(x_i)]$) is the fitted value; $F^*(x_i) = \frac{i}{n+1}$ is the empirical distribution; $F(x_i)$ is the theoretical distribution and D_n is the maximum value of the absolute difference between the empirical distribution and the theoretical distribution. Here, the confidence degree of K_f was set as 5%, and the samples are thought to obey the theoretical distribution when $K_f < 1.35$. The results of parameters estimation are shown in Table 3 and the probability density curves of the sample data and the fitting wind distribution for the station are shown in Fig. 3.

In this study, the basic wind speed U_{10} was determined as 36.5 m/s by combining the Chinese Standard (2004) and the research results. The design wind speed of 46.1 m/s at the deck height was determined.

2.2 Modal Parameters

A three-dimensional finite element model (FEM) of the self-anchored suspension bridge has been established in ANSYS to calculate the modal parameters, which include the natural frequencies and mode shapes of the bridge. The bridge deck, tower and pier are modeled by beam4 elements. The main cables and hangers are simulated by link10 elements. The bridge deck

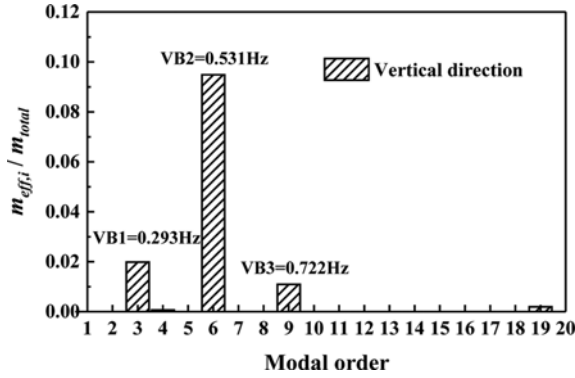


Fig. 4. Ratio of the Effective Mass to the Total Mass in the Vertical Direction for the First 20 Modes

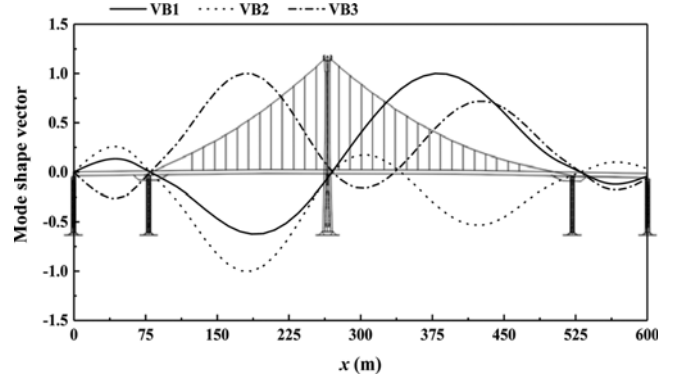


Fig. 5. Vertical Modal Shape Vector According to Mode

pavement and other accessory equipment are modeled by mass21 elements. The longitudinal bearing restraint of the deck is simulated by link8 elements. The model consists of 894 nodes and 1198 elements. Only the lower vertical bending (VB) modes are considered in further discussions.

The order number of modes was taken as 300 for the modal analysis in ANSYS. For the first 300 modes, the sum of the ratio of effective mass for the i^{th} mode $m_{\text{eff},i}$ in the vertical direction to the total mass m_{total} reached 0.925 and the 6th mode presented the maximum ratio 0.095 in the vertical direction. $m_{\text{eff},i} / m_{\text{total}}$ in the vertical direction for the first 20 modes of the whole bridge was presented, as shown in Fig. 4. From the Fig. 4, the first three VB modes of the bridge were identified.

The equivalent mass and the equivalent mass moment of inertia of the deck for the i^{th} mode, $m_{\text{eq},i}$ and $I_{\text{eq},i}$, which are used for the target value in the design of the section model for wind tunnel tests are given by (Hwang *et al.*, 2019):

$$m_{\text{eq},i} = \frac{\int_D m \phi_i^2(x) dx}{\int_D \phi_{vi}^2(x) dx} \quad (7)$$

$$I_{\text{eq},i} = \frac{\int_D I \phi_i^2(x) dx}{\int_D \phi_{vi}^2(x) dx} \quad (8)$$

where $\int_D$ means that integration is conducted only along the deck members, m and I are the mass and mass moment inertia of the members in a unit length respectively, $\phi_i(x)$ is the mode

shape vector of the i^{th} mode for all degrees of freedom in the FEM model, and $\phi_{vi}(x)$ is the mode shape vector of the i^{th} mode that is only considered for the translational degrees of freedom in the deck model, $\phi_{ri}(x)$ is the mode shape vector of the i^{th} mode that is only considered for the rotational degrees of freedom in the deck model. Table 4 summarizes the natural frequencies and the equivalent mass for the dominant three VB modes of the bridge deck. And, Fig. 5 presents the vertical mode shapes of the bridge, as obtained from three-dimensional FEM model analysis.

3. Expressions of VIV Response and Corresponding Performance Functions

3.1 Expressions of VIV Response

In design codes of different countries, such as Chinese Standard (2004), the allowable amplitude value (vertical bending or torsional direction) of the VIV response is clearly defined. During the wind-resistant design stage for bridges, the maximum VIV amplitude must be less than the allowable value. The expressions of the maximum VIV amplitude can be obtained as follows based on the basic assumptions of standard harmonic vortex-induced load and structural harmonic response:

3.1.1. Amplitude Expression y_l

The expression of y_l is obtained based on an single-degree-of-freedom model, shown in Eq. (9), without considering vortex moving as (Simiu and Scanlan, 1996):

$$m(\ddot{y} + 4\pi\zeta f_n \dot{y} + 4\pi^2 f_n^2 y) = \frac{1}{2} \rho U^2 B \cdot C_v \cdot \sin(2\pi f_s \cdot t) \quad (9)$$

$$y_l = \frac{C_v}{4\pi S_i^2 S_c} \cdot D \quad (10)$$

$$S_i = \frac{f_n \cdot D}{U} \quad (11)$$

$$S_c = \frac{4\pi m \zeta}{\rho B D} \quad (12)$$

where C_v is the vortex-induced aerodynamic force coefficient obtained based on the wind tunnel test fitting; S_i is the Strouhal

Table 4. Modal Parameters for VB Modes from FEM Analysis Model in Completion Stage

Mode	Natural frequency / Hz	Equivalent mass $m_{\text{eq},i}$ in vertical direction/(kg/m)	$\frac{m_{\text{eq},i}}{\sum m_{\text{eq},i} + \sum I_{\text{eq},i}}$
VB1	0.293	41.3×10^3	0.999
VB2	0.531	40.5×10^3	0.999
VB3	0.722	40.2×10^3	0.999

Note: $\sum m_{\text{eq},i} + \sum I_{\text{eq},i}$ means the total equivalent mass and equivalent mass moment of inertia for the i^{th} mode in the vertical, lateral and span directions of the deck.

number; S_c is the Scruton number; f_n and f_s are the structural natural fundamental frequency and the vortex shedding frequency, respectively; U is the mean velocity of the uniform flow; m is the mass per unit length of bridge deck; ζ is the damping ratio; ρ is the air density; B is the along-wind dimension of the bridge deck and D is the across-wind dimension of the bridge deck.

3.1.2 Amplitude Expression y_{II}

The expression of y_{II} is obtained based on the Ruscheweyh model (Ruscheweyh, 1996), which considers the span-wise correlation of vortex-induced forces and is written to Euro Code (European Standard, 2005) as:

$$y_{II} = \frac{C_v}{S_t^2 S_c} \cdot K \cdot K_w \cdot D \quad (13)$$

where K ($0.1 \leq K \leq 0.14$) is the mode shape factor; and K_w ($0.1 \leq K_w \leq 0.6$) is the effective correlation length factor.

3.1.3 Amplitude Expression y_{III}

y_{max} is obtained based on an single-degree-of-freedom model with consideration of the vortex moving along the deck, shown in Eq. (14), which can be used to simulate the locked vibration of the bridge deck with high fidelity as (Li *et al.*, 2018; Li, 2004):

$$m(\ddot{y} + 4\pi\zeta f_n \dot{y} + 4\pi^2 f_n^2 y) = \frac{1}{2} \rho U^2 B \cdot C_v \cdot \sin(2\pi\sqrt{f_s \cdot f_m} \cdot t) \quad (14)$$

$$y_{max} = \frac{C_v}{4\pi S_c} \cdot \left(\frac{U}{f_n D}\right)^2 \cdot D \quad (15)$$

$$L_l = \frac{U_e}{U_b} \quad (16)$$

$$S_t = \frac{f_n D}{U_b} \cdot \frac{2}{(1 + L_l)} \quad (17)$$

where, L_l is the lock-in range parameter defined as the ratio of the start wind speed U_b to the end wind speed U_e of the lock-in wind speed range $[U_b, U_e]$; f_m is the reduced frequency of moving vortex (Li *et al.*, 2018). When $f_m = f_s = f_n$, the VIV amplitude reaches the maximum value. Then, y_{III} can be expressed as:

$$y_{III} = \frac{C_v}{\pi S_t^2 S_c (1 + L_l)^2} \cdot D \quad (18)$$

3.2 Performance Functions of VIV Response

Chinese Standard (2004) gives rules for the VIV amplitude h_c or θ_c (corresponding to vertical or torsional direction) for the main girder of a long-span bridge as follows:

$$h_c < [h_a] = \frac{0.04}{f_b} \quad (19)$$

$$\theta_c < [\theta_a] = \frac{4.56}{B \cdot f_t} \quad (20)$$

In which, $[h_a]$ or $[\theta_a]$ is the allowable value (vertical bending or torsion) of the VIV amplitude; f_b or f_t is the vertical bending or torsion frequency of the bridge, respectively; and B is the along-wind dimension of the bridge deck. If the maximum VIV amplitude is beyond the allowable value, it indicates that the VIV amplitude is in the failure state. The performance function for the VIV amplitude of bridges can be obtained as:

$$g(y_{max}, [y_a]) = [y_a] - y_{max} \quad (21)$$

where $y_{max}(y_I, y_{II}$ or $y_{III})$ is the maximum VIV amplitude of bridges; $[y_a]$ represents $[h_a]$ or $[\theta_a]$. When $g(y_{max}, [y_a])$ is less than zero, it indicates that the maximum VIV amplitude is in the failure state. The corresponding performance functions of VIV response are shown as follows:

$$Z_I(y_I, [y_a]) = [y_a] - \frac{1}{4\pi} \cdot \frac{C_v}{S_t^2 S_c} \cdot D \quad (22)$$

$$Z_{II}(y_{II}, [y_a]) = [y_a] - (K \cdot K_w) \cdot \frac{C_v}{S_t^2 S_c} \cdot D \quad (23)$$

$$Z_{III}(y_{III}, [y_a]) = [y_a] - \frac{1}{\pi(1 + L_l)^2} \cdot \frac{C_v}{S_t^2 S_c} \cdot D \quad (24)$$

The theoretical result for the model parameters C_v , L_l and S_t in Eqs. (22)–(24) is currently not obtained. Hence, they are typically obtained based on the wind tunnel tests for a semi-empirical solution. The exceeding probability of VIV response is denoted by P_f , which will be calculated by using the SOFM method and verified based on the MCS method in this study.

3.3 Uncertainty of Model Parameters for VIV Response

The mean value of the model parameters is usually defined as the mean value of sample data of parameters or the expected value, which is within expected and acceptable limits. Uncertainties

Table 5. Uncertainties of the Model Parameters for Bridge VIV Response

Parameter	Distribution type	COV	Literature sources
Wind speed	Extreme value type I	0.15	Kareem, 1988b; Simiu and Scanlan, 1996
Air density	Normal	0.05	Assumption
Fundamental natural frequency	Lognormal	0.15	Kareem, 1988b
Structural damping	Lognormal	0.40	Davenport and Hill-Carroll, 1986
Geometrical dimension	Normal	0.10	Kareem, 1988b
Mass	Normal	0.10	Kareem, 1988b
Aerodynamic force coefficient	Lognormal	0.15	Kareem, 1988b;
Strouhal number S_t	Lognormal	0.15	Ostenfeld-Rosenthal <i>et al.</i> , 1992

in the model parameters for bridge VIV response come from the variations of flow velocity, across-wind dimension of the body, lift force coefficients, Strouhal number, Scruton number, etc. The statistical analysis of these parameters was based on the meteorological data, or the stochastic analysis using perturbation, second-moments, neural network, MCS, stochastic finite element methods, or obtained from wind-tunnel measurements available in the literature (Kareem, 1988a; Kareem, 1988b; Kwon *et al.*, 2015; Larsen *et al.*, 2000). In this study, the mean value of the model parameters are the employed distribution types of the model parameters and their coefficients of variation (COVs) were conservatively considered based on the summarization of the existing achievements, shown in Table 5.

4. Probabilistic Evaluation of VIV Response

In the section model wind tunnel tests with a scale of 1:25, the completion bridge stage, with three attack angles of -3° , 0° , and $+3^\circ$ was considered here. The absolute relative errors for the model frequency were less than 5%, and the mode damping ratios were 5‰ here. The vertical VIV displacements for the prototypical bridge deck were yielded from four laser displacement systems, the time history for the measurements of which in a certain time was shown in Fig. 6(a) and Fig. 6(b) presents the power spectrum density of the data, which identifies the frequencies for the VB modes. The vertical bending VIV of the bridge deck was detected in one lock-in wind speed interval in the low-turbulence intensity flow field, as shown in Table 6 and Fig. 7.

It should be noted that VIV amplitudes at attack angles of -3° and $+3^\circ$ conditions exceed the allowable value according to the Chinese Standard (2004). Moreover, the maximum single peak amplitude appeared at attack angle of $+3^\circ$. The major reason is that the aerodynamic force at attack angle of $+3^\circ$ is larger than those at attack angles of -3° and 0° . It also indicates that the vortices attached to the surface of the section and the vortex shedding modes at different attack angles are different. The influence degree of vortices attached to the surface of the section on the vibration response at the attack angle of $+3^\circ$ is larger than the others for this bridge.

The empirical distribution types and the statistical properties of

Table 6. Single Peak Amplitude and Corresponding Lock-in Wind Speed of Vertical VIV for the Deck

Wind attack angle /degree	Lock-in wind speed /(m/s)	Natural frequency / Hz	Single peak amplitude /m	Allowable amplitude [h_a] /m
+3	7.6 ~ 11.4	0.293	0.248	0.136
0	7.6 ~ 11.0	0.293	0.134	
-3	7.6 ~ 10.0	0.293	0.138	

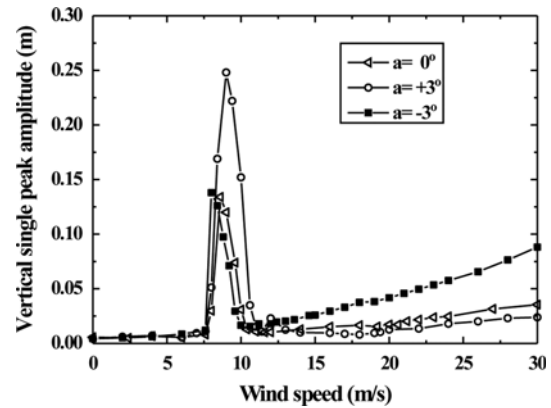


Fig. 7. VIV Vertical Single Peak Amplitude of Bridge Deck with Different Attack Angles

the random variables in Eqs. (22)–(24) for the maximum vertical bending amplitude are given in Table 7. In the calculations, S_f was obtained by the wake surveys; L_f and C_v were calculated based on the linear relationship of $L_f - S_c$ and $C_v - S_c$, which were derived from wind tunnel tests for various damping ratios; and both values of K and K_w were selected from the European Standard (2005).

The exceeding probabilities of the VIV response P_f were calculated using the SOFM method, which was verified by the MCS (with iteration of 10^6) method, as shown in Table 8. It is shown that probabilities P_f of VIV maximum amplitude exceeding the allowable amplitude are all greater than 50% and that at attack angle of $+3^\circ$ was the maximum, which exceeded 80%. The results obtained by the SOFM method were close to those based on MCS method, and those at attack angles of 0° and -3° were safer than those based on MCS method. Otherwise, the

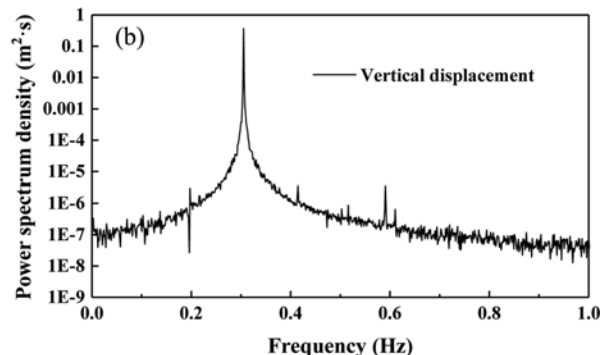
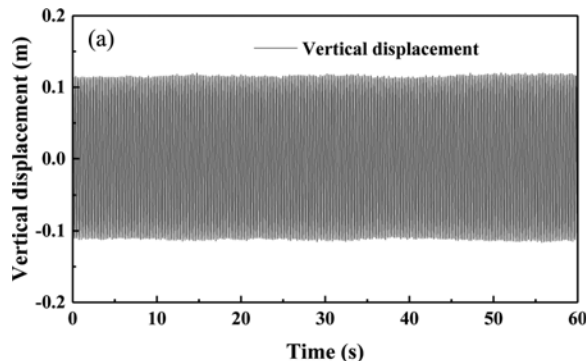


Fig. 6. Measurements of Vertical VIV Displacement in a Certain Time: (a) Time History of Vertical VIV Displacement, (b) Power Spectrum Density

Table 7. Experience Distribution Type and Statistical Properties of Random Variables

Item		Distribution type	Mean value		COV
Z_I	C_v	Lognormal	+3°	0.219	0.15
			0°	0.118	
			-3°	0.122	
Z_{II}			+3°	1.216	
			0°	0.657	
			-3°	0.677	
Z_{III}			+3°	0.140	
			0°	0.079	
			-3°	0.091	
S_I		Lognormal	0.1434		0.15
S_c		Normal	12.28		0.40
L_I	Extreme value I	+3°	1.500	0.10	
		0°	1.447		
		-3°	1.316		
K		Lognormal	0.11		0.10
K_w		Lognormal	0.13		0.10
D/m		Normal	3.6		0.10
$[y_a]/m$		Normal	0.136		0.10

Note: +3°, 0° and -3° are the attack angles of +3°, 0° and -3°, respectively

time-consuming based on the SOFM method is less than MCS method. This indicates that the SOFM method is high-precision and effective in the exceeding probability calculation of the bridge VIV response.

It should be noted that the value of the exceeding probabilities P_f based on above three performance functions Z_I , Z_{II} and Z_{III} are very close to each other even equal. One major reason is that the parameters were estimated based on the same group of dataset

(e.g., model properties and wind tunnel tests). On the other hand, it indicates that the common parameters of these three performance functions (e.g., S_c and S_I) have dominant influence on the VIV maximum amplitude. The corresponding sensitivity indexes of random variables were calculated and the results are presented in Table 9. It shows that the sensitivity indexes of S_c and S_I , which are quite sensitive to the mass and across-wind dimension, respectively, have dominant influence on the exceeding probabilities and the vortex-induced aerodynamic force coefficient C_v is the next. It is important and necessary to obtain more accurate and reliable information of these parameters.

5. Uncertainty Effect of Parameters

In reality, the uncertainty in the description of the parameters is hard to be described accurately because of many interfering factors and the lack of information. In order to investigate the uncertainty effect of parameters on the exceeding probability of VIV response, the different random nature of all parameters in the performance function Z_{III} are considered. In this study, the changes in the uncertainty of the parameters are adopted: 1) Increasing the values of the COV, the skewness and the kurtosis of all parameters to 1.5 times in increments of ten percent; 2) Changing the distribution types of the high sensitivity parameters of S_c and S_I . The corresponding exceeding probabilities are calculated, as shown in Figs. 8(a), 8(b), 8(c) and Table 10.

By changing the numerical characteristics of the parameters in the performance function Z_{III} , the exceeding probability of VIV response shows linear increasing with the increasing of the COV at both attack angles of 0° and -3° and the skewness at above three attack angles, but decreasing with that of the COV at the

Table 8. Exceeding Probability of VIV Maximum Vertical Bending Amplitude

Item	Attack angle	Z_I		Z_{II}		Z_{III}	
		SOFM	MCS	SOFM	MCS	SOFM	MCS
Time consuming/s	+3°	5.52	118.31	7.82	111.53	8.72	118.55
	0°	5.44	127.47	7.78	111.12	8.72	123.88
	-3°	5.44	128.34	7.96	111.10	8.70	120.73
$P_f/\%$	+3°	83.3	89.3	82.1	87.9	82.5	88.6
	0°	58.3	51.7	57.9	51.1	58.3	51.6
	-3°	60.2	54.1	59.5	53.3	60.0	53.8

Table 9. Sensitivity Indexes of Random Variables

Performance function	Attack angle	C_v	S_I	S_c	L_I	K	K_w	D	$[y_a]$
Z_I	+3°	0.32	-0.63	-0.64	—	—	—	0.22	-0.21
	0°	0.28	-0.55	-0.74				0.19	-0.19
	-3°	0.28	-0.56	-0.75				0.19	-0.19
Z_{II}	+3°	0.30	-0.60	-0.62	—	0.20	0.20	0.21	-0.20
	0°	0.27	-0.53	-0.72		0.18	0.18	0.18	-0.18
	-3°	0.27	-0.54	-0.71		0.18	0.18	0.18	-0.18
Z_{III}	+3°	0.31	-0.61	-0.64	0.21	—	—	0.21	-0.20
	0°	0.27	-0.54	-0.73	0.19			0.18	-0.18
	-3°	0.27	-0.54	-0.73	0.19			0.18	-0.18

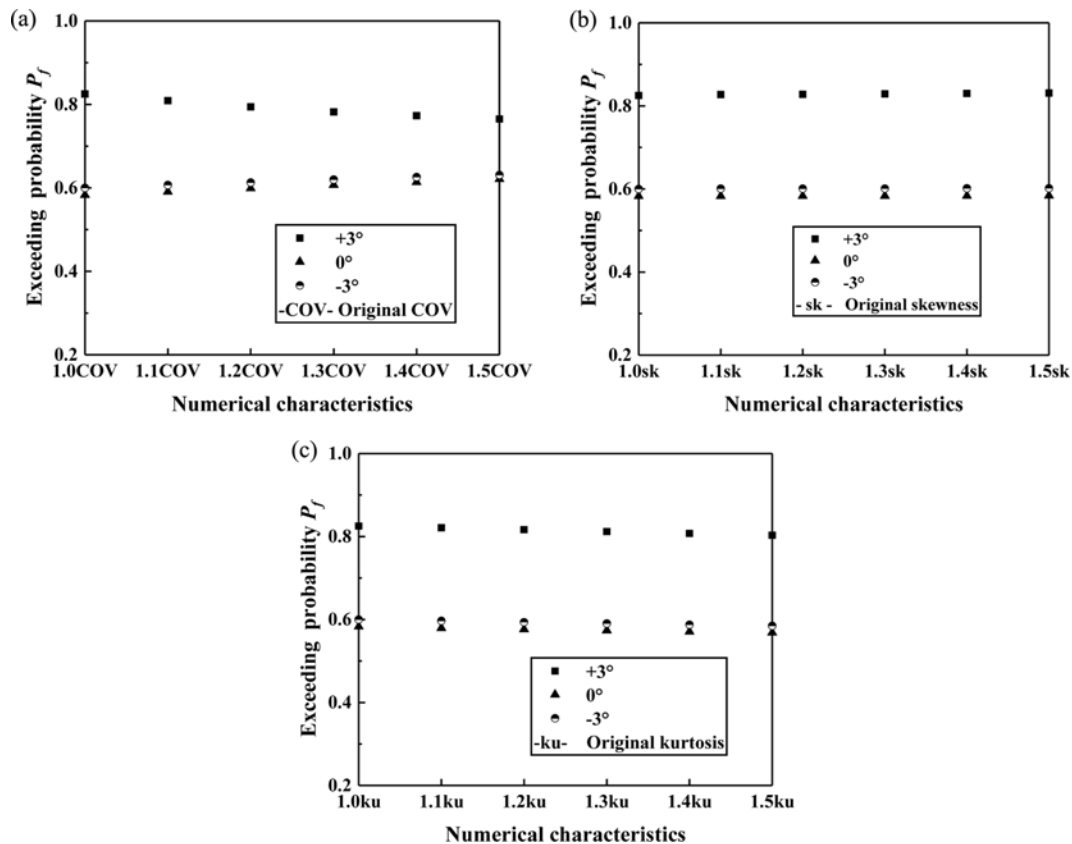


Fig. 8. Change Curve of the Exceeding Probability with the Numerical Characteristics of the Parameters: (a) COV, (b) Skewness, (c) Kurtosis

Table 10. VIV Exceeding Probability by Changing the Distribution Type of S_c and S_i

Parameter	Distribution type	$P_f / \%$					
		SOFM			MCS		
		+3°	0°	-3°	+3°	0°	-3°
S_c	Normal (original)	82.5	58.3	60.0	88.6	51.6	53.8
	Lognormal	82.6	58.3	60.0	89.6	55.0	57.2
S_i	Lognormal (original)	82.5	58.3	60.0	88.6	51.6	53.8
	Normal	82.6	58.3	60.0	88.4	51.1	53.3

attack angle of +3° and the kurtosis at three attack angles. A fifty percent increase in the COV and the skewness of parameters can raise the exceeding probability of VIV response by 6.5% and 0.7%, respectively. A fifty percent increase in the COV at the attack angle of +3° can reduce the exceeding probability of VIV response by 7.3%. And for the kurtosis, the reduction can reach 2.7%. It indicates that the effect of the COV and the kurtosis variations is more significant than that of the skewness. By only changing the distribution types of S_c and S_i , there is little change in the exceeding probability of VIV response based on the SOFM method. It showed that the exceeding probability of the VIV response is not sensitive to the distribution type of the parameters based on the SORM method. On the other hand, there is relatively more change based on the MCS method. The reason is that the MCS method results are calculated only based on the first two moments of the parameters.

6. Conclusions

A self-anchored suspension bridge was selected as an example to analyze the uncertainty influence on the probabilistic evaluation of VIV response. The probability density function of the wind speed at the deck level of the bridge was established. Three performance functions involving different considerations of the maximum response models of the bridge VIV were introduced in this study. The numerical characteristics (COV, skewness and kurtosis) and the distribution type in the parameter space was propagated to access its influence on the exceeding probability of the bridge VIV response utilizing the SOFM method. The conclusions are as follows:

1. The uncertainties of parameters involved in these performance functions have the varying degree influence to the exceeding probabilities of bridge VIV response, of which

Scruton number S_c and Strouhal number S_t have the dominant influence and the vortex-induced aerodynamic force coefficient C_v is the next. It is important and necessary to obtain more accurate and reliable information of these parameters.

2. The COV and the kurtosis variations of the parameters have more significant influence on the exceeding probability of the VIV response than the skewness for the bridge. The distribution type of parameters has little effect on the exceeding probability of VIV response based on the SOFM method, which has relatively more effect based on the MCS method. The exceeding probability of the VIV response is not sensitive to the distribution type of the parameter based on the SOFM method.

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